Léo Crampé

Los Angeles, CA | cramp@usc.edu | (805) 201-5238 | LinkedIn | Portfolio

Education

University of Southern California — **Los Angeles, CA**
Master of Science in Mechanical Engineering (Artificial Intelligence and Machine Learning Emphasis) — December 2027
Bachelor of Science in Biomedical Engineering (Mechanical Emphasis): **3.66 GPA** — May 2027
Relevant Coursework: Data Modeling, Simulation Methods, Statistics and Probability, Operations Management

Anglo-American University — **Prague, Czechia**
Study Abroad Program — August 2023 - December 2023

Experience

NeuroSense — **Los Angeles, CA**
Co-Founder — September 2025 - Present

- Engineered a portable vibration threshold testing device utilizing Arduino electronics, linear actuating motors, CAD, and 3D printing to objectively measure sensory loss in patients with peripheral neuropathy
- Advanced NeuroSense to the USC Viterbi Min Family Challenge semifinals, selected among 11 teams from 88 competitors and competing for $50,000 in startup funding
- Conducted 7 clinical stakeholder interviews with physicians, researchers, and healthcare leaders to validate unmet clinical needs and translate user feedback into device requirements and product strategy

Zimmer Biomet — **Montreal, QC**
Systems Design Intern — May 2025 - August 2025

- Tested the ROSA Knee System as part of the performance team focused on biomechanics analysis for cut accuracy evaluations, bone registration, knee state angles, and bone painting in accordance with ISO 13485 QMS and design validation requirements
- Reduced processing times from 3 minutes to 3 seconds across 200+ executions by developing a SolidWorks macro that automated alignment of femur and tibia models using 3D spatial coordinate algorithms for precise validation of robot calculated cut planes
- Authored sections of 7 FDA compliant DVeP & DVeRs for medical device design, gaining experience in regulatory compliant documentation and traceability processes in a french speaking engineering environment

Insead Business School for the World — **Fontainebleau, France**
Chief Operating Officer during Summer Program — July 2022 - August 2022

- Led as the Chief Operating Officer of a 6 member team in a simulated bicycle company, streamlining manufacturing by reducing design cost by 30% and implementing risk analysis and mitigation strategies to ensure product safety
- Established bold manufacturing and rebranding changes to differentiate ourselves from competitors and drive innovation

Projects

Computational Modeling of Action Potentials — **Los Angeles, CA**
Student Researcher — September 2025 - November 2025

- Characterized system behavior across 50+ simulations by developing an Excel and MATLAB based computational models of neuronal action potentials, analyzing voltage propagation, spike frequency, and conductance behavior under varying excitation and inhibition inputs to investigate system performance and variability

Electric Guitar — **Los Angeles, CA**
Electrical Design Lead — March 2025 - May 2025

- Developed and fabricated a custom electric guitar with an integrated active filter circuit to enhance tonal performance
- Designed and tested analog filters using op-amps and a charge pump for dual-voltage operation
- Created schematics and PCB layouts in KiCad, managing end-to-end circuit board production

Smart Knee Brace — **Los Angeles, CA**
CAD Design Lead — January 2025 - February 2025

- Constructed an Arduino-powered smart knee brace utilizing Fusion360 to assist athletes in recovering from patellar tendonitis 2x faster by integrating vibration therapy and real-time planar knee motion-tracking using accelerometers
- Implemented haptic feedback alerts to notify users of risky movements, promoting safer rehabilitation

Skills & Extracurriculars

Technical Skills: Solidworks, Fusion360, Siemens NX, AutoCAD, Autodesk Fusion, PTC Creo, CATIA V5, ICIDO, ENOVIA, Revit, TinkerCAD, MATLAB, Python, C/C++, Java, Arduino IDE, Labview, KiCad, LTspice, Microsoft Excel, Minitab, DFMEA, RCA, GMP, QA, ISO 13485 QMS, GD&T, FEA, DFMA, 3D Printing, PCR, Digital signal processing, Soldering, Welding, Oscilloscope, Strain gauge, Load cell, Accelerometer, Thermocouple, Pitot tube, FTIR, Gyroscope
Languages: Fully Bilingual in English and French
Organizations: Theta Tau Professional Engineering Society, Engineers Without Borders, MEDesign, Ski Team, Brazilian Jiu Jitsu
International Exposure: US and French Citizen, raised in Singapore; studied/worked in the U.S., Canada, France, and Czechia